Parallel and Distributed Skin Tone Analysis for Automated Foundation Shade Matching

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Abstract. *Selecting an appropriate cosmetic foundation shade can be challenging, particularly in online environments where products cannot be physically tested. This work proposes a parallel and distributed computer vision system that estimates skin tone from facial images and recommends compatible foundation shades by mapping both users and products to the Monk Skin Tone Scale. The system also detects regions affected by pigmentation-altering conditions, including acne, rosacea, and vitiligo, reducing their influence on skin tone estimation. Parallel and distributed processing techniques are incorporated into the image-analysis pipeline to improve computational performance and scalability. The proposed approach integrates skin analysis, cosmetic shade matching, and high-performance computing into a unified recommendation system.*

1. Introduction

The selection of cosmetic foundation is strongly dependent on the compatibility between a product's shade and the user's skin tone. Although cosmetic manufacturers typically provide shade names, photographs, and digital color representations for their products, determining the most appropriate shade remains largely dependent on subjective visual comparison. This process becomes particularly challenging in online environments, where users cannot directly test products and where variations in illumination, camera characteristics, and image acquisition conditions may alter the apparent color of the skin. Computer vision techniques provide an opportunity to automate this process by extracting skin color information from facial images and mapping it to standardized skin tone representations.

Automated foundation matching presents two important challenges. First, facial skin does not necessarily exhibit uniform pigmentation: conditions such as acne, rosacea, and vitiligo can produce localized color variations that may not represent an individual's predominant skin tone and may consequently bias its estimation. Second, facial image analysis involves multiple computationally intensive operations, including image preprocessing, skin segmentation, condition detection, color extraction, and shade comparison. An effective solution must therefore distinguish representative skin regions from pigmentation-altering regions.

This work aims to develop a computer vision system capable of estimating a user's skin tone from a facial image and recommending compatible cosmetic foundation shades. The detected skin tone is associated with the Monk Skin Tone Scale (Monk,

2019), which serves as an intermediate representation between the user and the cosmetic product database. Foundation shades are collected with their commercial names and digital color values obtained from manufacturers' official websites and subsequently mapped to the Monk Scale. Additionally, the proposed system identifies facial regions affected by acne, rosacea, and vitiligo, allowing these regions to be excluded or treated separately during skin tone estimation. These detections are intended to improve image processing and cosmetic recommendation and do not constitute medical diagnosis.

The main contribution of this work is the integration of skin tone estimation, pigmentation-aware facial analysis, cosmetic shade matching, and parallel and distributed computing into a unified recommendation pipeline. Parallelism is explored by concurrently processing independent image regions and computational tasks, including the detection of different pigmentation-altering conditions, while the distributed architecture separates major stages of the pipeline, such as image processing, skin analysis, and foundation-shade matching, into independently executable components. This approach enables the project to evaluate not only the accuracy of the resulting cosmetic recommendations, but also the performance gains and scalability provided by parallel and distributed execution.

This paper is organized as follows. Section 2 presents related works concerning parallel and distributed image processing, skin tone estimation, and automated detection of skin conditions. Section 3 describes the proposed methodology, as well as the project timeline. The subsequent sections present the experimental results, discuss the obtained findings, and conclude the work.

2. Related Works

Rakhimov et al. (2026) compared two parallel programming frameworks, CUDA (running on GPU) and OpenMP (running on CPU), applied to the Canny Edge Detection algorithm, testing images of varying sizes (512×512, 1024×1024, and 3200×2400). Their results showed that CUDA significantly outperformed OpenMP, achieving a speedup of 40x over sequential processing, compared to an average speedup of 7x for OpenMP. This work highlights the performance benefits of parallel computing for Image Processing applications.

Afrah et al. (2008) proposed Hive, a novel vision processing system architecture based on a peer-to-peer (P2P) network. The authors aimed to fill a gap in computer vision middleware by facilitating communication between independent cross-platform modules via an extensible protocol, allowing modules to form a vision processing pipeline. To demonstrate the simplicity and flexibility of the proposed architecture, the authors developed a distributed face tracking system, successfully executed on machines running Windows XP and Ubuntu 7.10. This demonstrates the viability of modular, cross-platform architectures for building vision processing pipelines, a principle our work draws on when designing the distributed pipeline for face detection, region exclusion, and skin tone matching, albeit within a different distributed processing paradigm.

Mbatha et al. (2024) evaluated the ability of Convolutional Neural Networks to identify an individual's skin tone under different lighting conditions. The authors adopted the Monk Scale (Monk, 2019), developed by Google to address the underrepresentation of black and brown skin tones on the previously widespread Fitzpatrick Scale. The proposed

model achieved considerably lower LAB L^2 distance errors on lighter skin tones than on darker skin tones, indicating reduced accuracy for the latter. Averaged across all tested lighting conditions, the model achieved an accuracy of approximately 56%, 84%, and 97% at distance thresholds of ≤ 0.5, 1, and 2, respectively. This disparity in performance between lighter and darker skin tones underscores a common limitation in automated skin tone classification systems, and motivates the emphasis this work places on ensuring accurate and equitable foundation matching across the full range of skin tones.

Shifting the focus to the detection of skin conditions, Liu et al. (2020) developed a Deep Learning System (DLS) capable of differentiating between 26 conditions. In addition to analyzing images, this system also leveraged 45 other types of data that would be available to physicians in a teledermatology service, such as demographic information and medical history. The authors compared the DLS against board-certified dermatologists, with the DLS achieving an accuracy of 0.66 compared to 0.63 for the human diagnosticians. This shows that automated systems can achieve diagnostic accuracy comparable to that of trained specialists, supporting the feasibility of an automated pre-processing step to identify and exclude skin regions affected by pigmentation-altering conditions before performing skin tone analysis, as proposed in this work.

Together, these works highlight two complementary threads. Rakhimov et al. (2026) and Afrah et al. (2008) demonstrate the performance and flexibility gains achievable through parallel and distributed vision processing, while Mbatha et al. (2024) and Liu et al. (2020) show that automated skin analysis is feasible, but still limited by reduced accuracy for darker skin tones and imperfect condition detection. Our work aims to combine these threads into a parallelized, distributed pipeline for skin tone detection, applied to the accurate selection of cosmetic products.

3. Methodology

3.1. Image Processing

For identifying the relevant areas of image, the image passes through a pre-processing step, in which the image is converted from the RGB color space into the CIELAB color space, which more closely portraits how the human eye perceive the distance between colors. The skin is then segmented by using a special K-means implementation proposed by Islam et al. (2024), where the number of clusters is dynamically calculated. The skin-related data were carefully selected from existing databases to maximize inclusivity. Data representing specific skin conditions were gathered from databases and synthetically generated by introducing skin conditions into previously collected images. To find the skin tone, the system calculates the ITA of the average skin color, which is then mapped into the Monk Skin Scale.

3.2. Tone Matching

The foundation data used for tone matching was manually collected and organized in a spreadsheet, and unlabeled data were manually annotated. To find the respective foundation, the system uses KNN to identify the closest foundation tone for the Monk Scale's tone obtained.

3.3. Algorithm Optimization

Given the inherent computational cost of segmentation operations, clinical condition identification, and topological matching, the pipeline is architecturally optimized. In the realm of parallel computing, image processing exploits concurrent execution through the OpenMP (CPU-oriented) and CUDA (GPU-oriented) libraries. This data-level parallelism allows independent processes on the image to run simultaneously, promoting a substantial speedup over purely sequential implementations.

3.4. Workload Distribution

The proposed system is structured in a decentralized pipeline architecture. Instead of internally distributing the execution of a single algorithm, the workflow is partitioned into independent sequential stages (K-means segmentation, ITA calculation, and KNN classification). Each of these stages can be allocated and processed on distinct physical machines, called nodes. These nodes interact cooperatively, transferring the results of the preceding stage to the machine responsible for the subsequent stage via a direct Peer-to-Peer communication protocol, which ensures overall system scalability.

3.5. Project Timeline

Time	Goal	Description
Weeks 1–3	Database Structuring	Collection and organization of the database (brand, line, shade, name, link). Extraction of color values from official websites.
Weeks 4–5	Standardization & Scaling	Mapping of collected foundation shades to the ten representative points of the Monk Scale.
Weeks 6–8	Tone Recognition & Computer Vision	Development of facial segmentation, color extraction, dynamic K-means clustering, and targeted detection for acne, rosacea, and vitiligo.
Weeks 8–9	Parallelism and Distribution Integration	CUDA/OpenMP application for concurrent image detection. Visual pipeline modularization and P2P communication.
Weeks 10–11	Evaluation, Adjustments and Documentation	Accuracy measurement across Monk Scale extremes, speedup evaluation, scalable distributed execution, and technical documentation drafting.

4. Skin Detection

Since the input is user-dependent, the image used for the foundation matching process may contain elements that negatively impact the color analysis stage, such as eyes, hair, brows, glasses, or types of skin condition. For our purposes, these areas can be considered as noise, making it necessary to isolate the applicable areas of the skin from the rest of the image.

4.1. Pre-Processing

Before the skin is segmented, the image is first converted from the RGB color space, into the CIELAB color space. This step is crucial, since CIELAB separates lightness information (Luminance) from color information (Chrominance), allowing the identification of skin areas while avoiding the interference of the ambient lighting conditions of the captured image, resulting in larger skin clusters.

4.2. Segmentation

The process of separating the skin from the image relies on a special K-means implementation where the number of clusters is automatically defined by analyzing the properties of the objects present in the scene, as Islam et al. (2024) demonstrated. The process of region growing is used to recover the remaining skin pixel, guaranteeing that no relevant regions are omitted.

5. Tone Detection & Foundation Matching

After the isolated segmentation of healthy skin, tone detection is based on calculating the average color of the segmented region. From this average, the Individual Typology Angle (ITA) is calculated (1). The ITA value is generally mapped into the Fitzpatrick Tone Scale, but due to their shortcomings in both accuracy and inclusivity, the ITA value is mapped to the corresponding shade on the Monk Skin Tone Scale instead, which Ulrich et al. (2025) claims to be superior. Finally, to find the correspondent foundation, the system uses KNN, to identify which foundation tone is most closely related to the Monk Skin tone obtained.

$$\text{ITA} = [\arctan((L^* - 50)/b^*)] \cdot 180/\pi \tag{1}$$

ITA calculation formula

6. Pipeline

7. Results

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